

# Cyril Creque-Sarbinowski

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■ External Resource

■ Slides/Notes

■ Paper

## Education

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|---------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>Johns Hopkins University (JHU)</b><br><i>Ph.D in Physics &amp; Astronomy</i><br>Advisor: Marc Kamionkowski | <b>Baltimore, MD</b><br>2017-2022 |
| <b>Massachusetts Institute of Technology (MIT)</b><br><i>B.S. in Physics &amp; Mathematics</i>                | <b>Cambridge, MA</b><br>2013-2017 |

## Positions Held

|                                                                                       |                                                       |
|---------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Center for Computational Astrophysics (CCA)</b><br><i>Flatiron Research Fellow</i> | <b>New York, NY</b><br>August 2022-Present            |
| <b>New York University (NYU)</b><br><i>Visiting Scholar</i>                           | <b>New York, NY</b><br>September 2022- September 2024 |

## Awards, Scholarships, Fellowships

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|-----------------------------------------------------------------------|-----------|
| <b>National Science Foundation</b><br><i>Graduate Research Fellow</i> | 2019-2022 |
| <b>Gates Millenium Scholars Program</b><br><i>Scholar</i>             | 2013-2022 |
| <b>JHU Kelly Miller Fellowship</b><br><i>Fellow</i>                   | 2017-2019 |

## Teaching/Outreach

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|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Sound Plus Science</b><br><i>Chief Facilitator &amp; Mentor, Developed Curriculum [Sample]</i>                                                 | <b>Civic Hall</b><br>March 2024-Present   |
| <b>GEM Annual Conference</b><br><i>Flatiron Institute Booth Staff</i>                                                                             | September 2023, 2024                      |
| <b>ASTRO 10000: Basic Concepts in Astronomy</b><br><i>Guest Lecturer</i>                                                                          | <b>Hunter College</b><br>May 2023         |
| <b>PHYS 111: Introductory Physics</b><br><i>Virtual Guest Lecturer</i>                                                                            | <b>Moravian University</b><br>Spring 2022 |
| <b>PHYS2300: Quantum Theory of Fields I</b><br><i>Guest Lecturer</i>                                                                              | <b>Brown University</b><br>Spring 2022    |
| <b>AS.171.107 General Physics for Physical Science Majors</b><br><i>Teaching Assistant</i><br>Lecturer: Prof. Robert Leheny & Prof. Rosemary Wyse | <b>JHU</b><br>Fall 2017                   |
| <b>AS.173.111 General Physics Laboratory I</b><br><i>Section Head</i>                                                                             | <b>JHU</b><br>Fall 2017                   |

## Selected Conferences, Workshops, Seminars

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\* = *Invited*, † = *Organizer*

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|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 2024.....           |                                                                                                                                      |               |
| Workshop            | Cosmology in the Adriatic – From PT to AI,<br>BCCP, Ruder Bošković Institute, Faculty of Science in Split                            | July          |
| Seminar*            | Cosmology Group Meeting, Perimeter Institute                                                                                         | July          |
| Program             | Dark Matter Theory, Simulation, & Analysis in the Era of Large Surveys,<br>Kavli Institute for Theoretical Physics (KITP), Affiliate | May-June      |
| Workshop            | The Future of Dark Matter, Columbia University/Flatiron Institute                                                                    | April         |
| Symposium†          | Caribbean Future of Science Symposium                                                                                                | March         |
| Seminar*            | KICP Seminar, University of Chicago                                                                                                  | January       |
| 2023.....           |                                                                                                                                      |               |
| Workshop*           | Large-Scale Parity Violation Workshop,<br>Academia Sinica Institute of Astronomy & Astrophysics (ASIAA)                              | December      |
| Workshop†           | Parity Violation from Home                                                                                                           | October       |
| Talk*               | Cosmology Talks                                                                                                                      | August        |
| Workshop*           | Ending Inflation & the Hot Big Bang, Stonybrook University                                                                           | June          |
| Workshop*           | Modified Gravity Program,<br>Illinois Center for Advanced Studies of the Universe                                                    | May           |
| Symposium*          | Future Faculty Symposium in Physics, Princeton University                                                                            | April         |
| Seminar*            | Quantum Gravity Seminar, Perimeter Institute                                                                                         | April         |
| Seminar*            | Cosmology Seminar, Max Planck Institute for Astrophysics                                                                             | April         |
| Seminar*            | Cosmology Seminar, Tufts/MIT                                                                                                         | February      |
| Seminar*            | Brown University IDEA Seminar                                                                                                        | February      |
| 2022.....           |                                                                                                                                      |               |
| Conference          | NSBP Annual Conference                                                                                                               | November      |
| Workshop*           | Guest Expert, Cosmology Talks Parity-Violation Mini-Workshop                                                                         | November      |
| Seminar*            | Theory Group Seminar, Imperial College London                                                                                        | November      |
| Talk*               | AstroParticle Lunch, The Ohio State University                                                                                       | September     |
| Talk                | Early Universe/Cosmology Lunch Discussion,<br>Princeton University/Institute of Advanced Study                                       | September     |
| Workshop†           | NeXus Workshop, CCA, Flatiron Institute                                                                                              | August        |
| Workshop            | Large-Scale Structure Cosmology beyond 2-Point Statistics,<br>Aspen Center for Physics                                               | May-June      |
| Seminar*            | Brown University                                                                                                                     | March         |
| $t \leq 2021$ ..... |                                                                                                                                      |               |
| Workshop            | NeXus Workshop, CCA, Flatiron Institute                                                                                              | June 2021     |
| Conference          | Phenomenology 2021                                                                                                                   | May 2021      |
| Workshop            | XIX International Workshop on Neutrino Telescopes                                                                                    | February 2021 |
| Conference          | Neutrino 2020                                                                                                                        | June 2020     |
| Seminar*            | Physics Seminar, Bowdoin College                                                                                                     | November 2019 |
| Conference          | NSBP Annual Conference                                                                                                               | November 2019 |
| Conference          | Cosmic Controversies,<br>Kavli Institute for Cosmological Physics (KICP)                                                             | October 2019  |
| Symposium           | Symposium in Honor of the Legacy of Vera Rubin,<br>Georgetown University                                                             | June 2019     |
| Workshop            | Novel Ideas for Dark Matter,<br>Princeton Center for Theoretical Science (PCTS)                                                      | January 2019  |
| Conference          | NSBP Annual Conference                                                                                                               | November 2018 |

|            |                                                                                               |               |
|------------|-----------------------------------------------------------------------------------------------|---------------|
| Conference | Identification of Dark Matter, Brown University                                               | July 2018     |
| Workshop   | Second Annual Line-Intensity Mapping Workshop (IM@Hopkins),<br>Johns Hopkins University (JHU) | July 2017     |
| Conference | Neutrinos & Light Particles in Cosmology,<br>Berkeley Center for Cosmological Physics (BCCP)  | June 2016     |
| Conference | NSBP Annual Conference                                                                        | February 2015 |

## Professional Activities

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|-----------|---------------------------------------------------------------------------------------------------|---------------|
| Referee   | Phys. Rev. D (2023–), Phys. Rev. Lett. (2023–)<br>MNRAS (2022–), Philos. Trans. R. Soc. A (2024–) |               |
| Leader    | NeXus                                                                                             | 2022–Present  |
| Organizer | Particles Vs. New Probes (P Vs. NP), Lead Organizer                                               | March 2025    |
|           | Caribbean Future of Science Symposium, SOC                                                        | March 2024    |
|           | Parity Violation from Home, Co-Organizer                                                          | October 2023  |
|           | NeXus Workshop, Lead Organizer                                                                    | August 2022   |
| Review    | GEM Fellowship Program                                                                            | November 2023 |
| Panelist  |                                                                                                   |               |

## Publications [13 ref. journal pub. (10 1st-author) +1 submitted, 300+ citations, h-index: 8]

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- [1] S. Alexander, H. Bernardo, and **C. Creque-Sarbinowski**, “Nontriviality of dynamical Chern-Simons gravity and the standard model,” *Phys. Rev. D* **110** no. 2, (2024) 025015, [arXiv:2403.15657 \[hep-th\]](#).
- [2] S. Alexander, **C. Creque-Sarbinowski**, H. Gilmer, and K. Freese, “Higgs Inflation and the Electroweak Gauge Sector,” [arXiv:2306.04671 \[hep-ph\]](#).
- [3] J. Flitter, **C. Creque-Sarbinowski**, M. Kamionkowski, and L. Dai, “Magnetic fields from compensated isocurvature perturbations,” *Phys. Rev. D* **107** no. 10, (2023) 103536, [arXiv:2304.03299 \[astro-ph.CO\]](#).
- [4] **C. Creque-Sarbinowski**, S. Alexander, M. Kamionkowski, and O. Philcox, “Parity-violating trispectrum from Chern-Simons gravity,” *JCAP (20th Anniversary Special Issue)* **11** (2023) 029, [arXiv:2303.04815 \[astro-ph.CO\]](#).
- [5] B. Zhou, L. Reali, E. Berti, M. Çalışkan, **C. Creque-Sarbinowski**, M. Kamionkowski, and B. S. Sathyaprakash, “Subtracting compact binary foregrounds to search for subdominant gravitational-wave backgrounds in next-generation ground-based observatories,” *Phys. Rev. D* **108** no. 6, (2023) 064040, [arXiv:2209.01310 \[gr-qc\]](#).
- [6] B. Zhou, L. Reali, E. Berti, M. Çalışkan, **C. Creque-Sarbinowski**, M. Kamionkowski, and B. S. Sathyaprakash, “Compact Binary Foreground Subtraction in Next-Generation Ground-Based Observatories,” [arXiv:2209.01221 \[gr-qc\]](#).
- [7] S. Alexander and **C. Creque-Sarbinowski**, “UV completions of Chern-Simons gravity,” *Phys. Rev. D* **108** no. 10, (2023) 104046, [arXiv:2207.05094 \[hep-ph\]](#).
- [8] **C. Creque-Sarbinowski**, J. Hyde, and M. Kamionkowski, “High-energy astrophysical neutrinos from cosmic strings,” *Phys. Rev. D* **107** no. 12, (2023) 123019, [arXiv:2206.06377 \[hep-ph\]](#).
- [9] **C. Creque-Sarbinowski**, M. Kamionkowski, and B. Zhou, “Seeking neutrino emission from AGN through temporal and spatial cross-correlation,” *Phys. Rev. D* **105** no. 12, (2022) 123035, [arXiv:2111.08012 \[astro-ph.HE\]](#).

- [10] **C. Creque-Sarbinowski**, M. Kamionkowski, and B. Zhou, “AGN variability in the age of VRO,” *Astrophys. J.* **941** (2022) 41, [arXiv:2110.13149 \[astro-ph.GA\]](#).
- [11] **C. Creque-Sarbinowski**, J. Hyde, and M. Kamionkowski, “Resonant neutrino self-interactions,” *Phys. Rev. D* **103** no. 2, (2021) 023527, [arXiv:2005.05332 \[hep-ph\]](#).
- [12] **C. Creque-Sarbinowski**, L. Ji, E. D. Kovetz, and M. Kamionkowski, “Direct millicharged dark matter cannot explain the EDGES signal,” *Phys. Rev. D* **100** no. 2, (2019) 023528, [arXiv:1903.09154 \[astro-ph.CO\]](#).
- [13] **C. Creque-Sarbinowski** and M. Kamionkowski, “Searching for Decaying and Annihilating Dark Matter with Line Intensity Mapping,” *Phys. Rev. D* **98** no. 6, (2018) 063524, [arXiv:1806.11119 \[astro-ph.CO\]](#).
- [14] **C. Creque-Sarbinowski**, S. Bird, and M. Kamionkowski, “Cross-correlation between thermal Sunyaev-Zeldovich effect and the integrated Sachs-Wolfe effect,” *Phys. Rev. D* **94** no. 6, (2016) 063519, [arXiv:1606.00839 \[astro-ph.CO\]](#).

## References

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